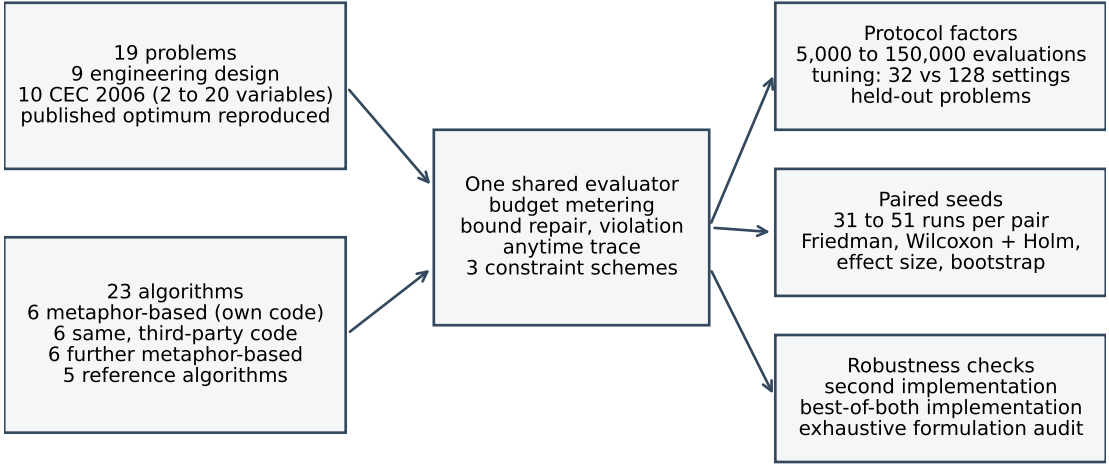
19 problems
9 engineering design
10 CEC 2006 (2 to 20 variables)
published optimum reproduced
23 algorithms
6 metaphor-based (own code)
6 same, third-party code
6 further metaphor-based
5 reference algorithms
One shared evaluator
budget metering
bound repair, violation
anytime trace
3 constraint schemes
Protocol factors
5,000 to 150,000 evaluations
tuning: 32 vs 128 settings
held-out problems
Paired seeds
31 to 51 runs per pair
Friedman, Wilcoxon + Holm,
effect size, bootstrap
Robustness checks
second implementation
best-of-both implementation
exhaustive formulation audit